Name: ______________________ Date: ______________________

SLIDE 1

NOTES

Class Agenda

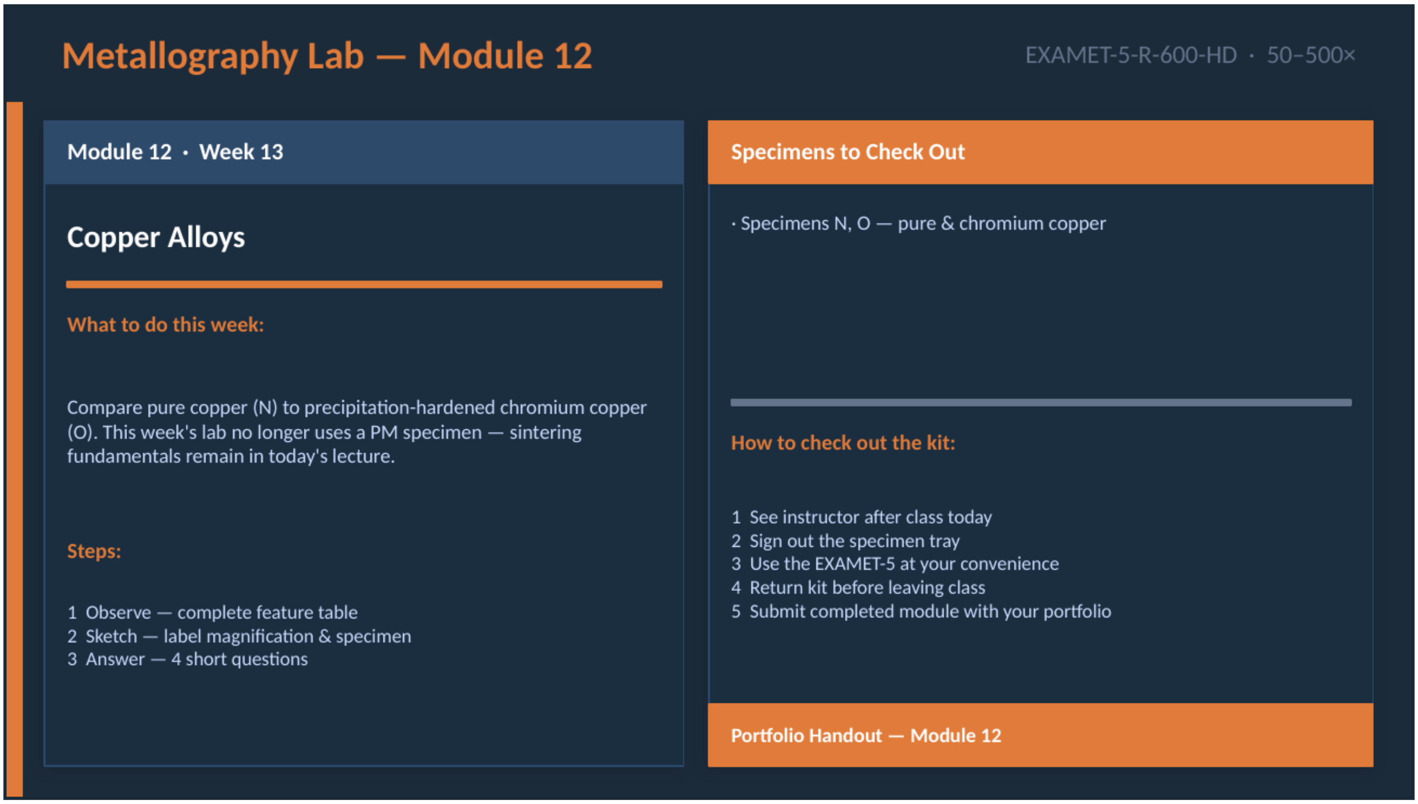
01 **Powder Production & Compaction**
Atomisation methods, particle characteristics, die compaction, density gradients, and the green compact

02 **Sintering & Advanced PM Processes**
Sintering stages, atmosphere control, liquid phase sintering, MIM, HIP, and additive manufacturing

03 **PM Defects & Process Selection**
Density gradients, lamination cracks, blistering — and the Unit 4 process selection framework

SLIDE 2 *Class Agenda*

NOTES



SLIDE 3 *Metallography Lab — Module 12 EXAMET-5-R-600-HD · 50–500×*

NOTES

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 ← here Nov 16 Powder Metallurgy	

Week 17 · FINAL EXAM

SLIDE 4 *MECH 191 — Course & Unit Roadmap*

NOTES

Week 13 — At a Glance

Topics

Powder production — water atomisation, gas atomisation, oxide reduction, electrolytic, and carbonyl processes; particle characterization
Compaction — blending with lubricants, die compaction, density gradients, green compact strength, and isostatic pressing (CIP/HIP)
Sintering — atmosphere control, three stages (burn-off, neck formation, densification), and liquid phase sintering
Advanced PM — metal injection molding (MIM), hot isostatic pressing, and additive manufacturing with metal powder feedstock
PM defects — density gradients, lamination cracks, blistering, and dimensional variation from sintering shrinkage
Process selection — matching alloy, geometry, tolerance, volume, and cost requirements across all Unit 4 processes

Resources

Reference Material

Kalpakjian Ch. 17
Powder Metallurgy — production, compaction, sintering, and applications

Visualizers

MPIF Process Overview · Sintering Animation
Atomisation & Powder Production
Manufacturing Process Selection

Key Fact

Conventional PM: 85–95% theoretical density · Sintering shrinkage: 1–3%

Next: Week 14 — Welding Processes

SLIDE 5

NOTES

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What is powder metallurgy and what advantages does it offer over casting and machining?
2. How are metal powders produced, compacted, and sintered?
3. What are the limitations of powder metallurgy and what defects should a technician look for?
4. How do you select the right manufacturing process for a given application?

SLIDE 6 *Core Questions*

NOTES

Powder Production — How Metal Powders Are Made

Production Methods

Water Atomisation

Molten metal stream broken by high-pressure water jets — irregular, oxidized particles, 50–200 µm. Most widely used for iron and steel; low cost; requires reducing atmosphere sintering to remove surface oxides.

Gas Atomisation

Inert gas (argon or nitrogen) jets produce spherical, clean particles. Higher cost; preferred for stainless steel, nickel superalloy, titanium, and tool steel. Better flowability and packing density than water atomised powder.

Oxide Reduction

Iron oxide reduced by hydrogen or CO at elevated temperature — sponge iron with irregular, porous morphology. Excellent green strength from mechanical particle interlocking. Widely used for structural iron PM parts.

Electrolytic & Carbonyl

Electrolytic deposition produces high-purity dendritic copper and iron. Carbonyl process decomposes metal carbonyls to yield extremely fine (1–10 µm) spherical iron or nickel powder — used in MIM and speciality applications.

Powder Characterization

Powder characteristics directly control compaction behavior, green density, and sintered properties — they are controlled specifications, not incidental measurements.

Particle size distribution

Controls green density and flow — finer particles pack better but flow poorly; measured by sieve analysis or laser diffraction.

Particle shape

Irregular particles interlock for higher green strength; spherical particles flow better. Shape is set by the production method.

Apparent & tap density

Apparent density: freely poured; tap density: after vibration. Hausner ratio = tap/apparent — indicates flow and compressibility.

Flow rate

Measured by Hall flowmeter — critical for automatic die filling; poor flow causes density variations across the compact.

SLIDE 7 *Powder Production — How Metal Powders Are Made*

NOTES

Compaction — Die Filling, Pressing & the Green Compact

Die Compaction Process

1 Blending

Powder blended with lubricant (zinc stearate) to reduce die wall friction and aid ejection; alloying additions (graphite for carbon content, copper for strengthening) are mixed in uniformly.

2 Die Filling

Powder gravity- or force-fed into a closed die. Uniform filling is critical — inconsistent fill causes green density variation and dimensional scatter in sintered parts.

3 Pressing — 150 to 700 MPa

Upper and lower punches press simultaneously — pressing from both ends reduces the density gradient through the part height. Pressure proportional to final density.

4 Ejection — Green Compact

Compact ejected from the die. Green strength 5–15 MPa from mechanical particle interlocking — fragile; handled carefully before sintering. Predictable 1–3% shrinkage on sintering.

Density Gradient & Isostatic Pressing

The Density Gradient Problem

Compaction pressure attenuates with distance from the punch faces due to inter-particle and die-wall friction — tall parts have lower density in the center. Controlled by using multiple independently moving punch segments and limiting the height-to-diameter ratio. Low-density zones have lower strength and become failure initiation sites in service.

Isostatic Pressing — Eliminating Gradients

Powder sealed in a flexible membrane compressed by fluid pressure from all directions simultaneously — no density gradient by design.

CIP — Cold Isostatic Pressing

Room temperature; used for large or complex shapes where die compaction cannot reach uniform density.

HIP — Hot Isostatic Pressing

Simultaneous heat and pressure (100–200 MPa argon) — 99%+ theoretical density; properties approach wrought. Used for aerospace superalloy discs, titanium components, and tool steel dies.

SLIDE 8 *Compaction — Die Filling, Pressing & the Green Compact*

NOTES

Sintering — Densification Without Melting

Three Stages of Sintering

1 **Burn-off — 300 to 500°C**

Lubricant is volatilised and removed from the compact before the sintering temperature is reached. This stage cannot be rushed — trapped lubricant vapour expands and causes surface blistering.

2 **Neck Formation — 70 to 90% of Tm**

At sintering temperature, atoms diffuse across particle contact points forming necks. Porosity is reduced rapidly and strength increases. For iron PM: approximately 1100–1150°C.

3 **Densification & Grain Growth**

Continued sintering reduces porosity further and grains coarsen. Densification is desirable; grain growth reduces mechanical properties — time and temperature must be optimised for the alloy.

Atmosphere & Liquid Phase Sintering

Sintering Atmosphere

Atmosphere must prevent oxidation throughout the furnace cycle.

Hydrogen / Dissociated ammonia
Iron, copper, and their alloys — reducing atmosphere removes surface oxides from the powder particles.

Nitrogen-hydrogen mixtures
Most structural PM steels — good reducing power at lower cost than pure hydrogen.

Vacuum
Stainless steel, titanium, and reactive metals — prevents any oxidation at sintering temperature.

Liquid Phase Sintering

A small amount of a lower-melting additive (cobalt in WC-Co; copper in iron PM) melts during sintering and fills pores by capillary action — dramatically accelerates densification. This is the key to tungsten carbide-cobalt cutting tool manufacture: WC particles bonded by liquid cobalt that re-solidifies on cooling.

SLIDE 9 *Sintering — Densification Without Melting*

NOTES

Secondary Operations & PM Applications

Secondary Operations

Coining (Re-pressing)
Sintered parts re-pressed in a die to close surface porosity and improve dimensional accuracy — increases density by 2–5% in surface layers.

Sizing
Re-pressing to correct dimensional variation from sintering shrinkage — achieves tolerances of ±0.025 mm on critical dimensions.

Infiltration
Liquid copper or bronze drawn into pores by capillary action — fills porosity, increases density and strength, improves machinability.

Oil Impregnation
Vacuum-impregnated porous bronze or iron bearings — oil stored in interconnected pores and released to the bearing surface during operation.

Heat Treatment
PM parts through-hardened, case-carburised, or induction hardened — porosity slightly reduces hardenability; thin sections monitored carefully.

Major PM Applications

Automotive
Connecting rod bearings, valve seat inserts, cam lobes, transmission synchroniser hubs — largest PM market by volume.

Cutting Tools
Tungsten carbide-cobalt (WC-Co) inserts and drill blanks — PM via liquid phase sintering is the only practical route to WC-Co.

Self-Lubricating Bearings
Oil-impregnated porous bronze and iron bearings in appliances, office equipment, and automotive accessories.

Magnetic Components
Soft magnetic cores for motors and transformers; permanent magnets (NdFeB, SmCo) for electric motors and sensors.

Medical & Dental
Implant components and dental crowns produced by MIM or DMLS — complex geometry, high dimensional precision required.

SLIDE 10 *Secondary Operations & PM Applications*

NOTES

Advanced PM Processes — MIM, HIP & Additive Manufacturing

Metal Injection Molding (MIM)	Hot Isostatic Pressing (HIP)	Additive Manufacturing
Very fine powder (1–20 µm) mixed with a polymer binder to form a feedstock that is injection molded like a plastic part. Binder removed by solvent and thermal debinding; part then sintered to 96–99% density. Produces complex small parts under 100 g with good surface finish and tight tolerances — used for firearms components, medical instruments, orthodontic brackets, and electronic hardware. No die-compaction geometry limits.	Powder or pre-sintered part sealed in a metal can subjected to simultaneous high temperature and high isostatic gas pressure (100–200 MPa argon). Eliminates virtually all porosity — achieves 99%+ of theoretical density with isotropic properties approaching wrought material. Used for aerospace superalloy turbine discs, titanium structural components, and tool steel dies where 100% density is a design requirement.	SLM (selective laser melting), EBM (electron beam melting), and DED (directed energy deposition) use gas atomised metal powder as feedstock — fused layer by layer to build near-net-shape parts. No tooling required; complex internal channels and lattice structures impossible by conventional methods. Fastest growing area of PM technology — used for aerospace brackets, medical implants with porous ingrowth surfaces, and tooling inserts with conformal cooling.

SLIDE 11 *Advanced PM Processes — MIM, HIP & Additive Manufacturing*

NOTES

PM Defects — Recognition and Detection

Defect	Cause	Appearance	Location	Detection	Action
Density Gradients	Uneven compaction — pressure attenuates from punch faces due to inter-particle and die-wall friction; most severe in tall or complex parts	No surface indication; lower hardness on Vickers traverse across the compact section; lower density zones visible on cross-section	Interior of compact, away from punch contact faces; typically mid-height of tall parts	**Hardness traverse; density measurement; UT velocity mapping**	Multiple punch segments; limit height-to-diameter ratio; optimise lubricant content
Lamination Cracks	Spring-back of compact on die ejection — elastic recovery of the powder mass causes horizontal fracture planes perpendicular to the pressing direction	Thin planar separations perpendicular to pressing direction — may not be visible on the surface; crack plane parallel to the compact base	Through compact cross-section, perpendicular to pressing axis; often near the ends of the compact	**UT — planar crack reflects beam clearly; visual on fracture surface if crack opens during handling**	Reduce ejection speed; increase die wall taper; optimise lubricant level and compaction pressure
Surface Blistering	Trapped lubricant vapour expands during sintering — caused by rushing the burn-off stage (300–500°C) or by contamination with low-vapour-pressure material in the powder blend	Raised blisters or bubbles on the sintered surface; 1–10 mm diameter depending on contamination severity	Surface of sintered compact; most common where lubricant burn-off is slowest in the furnace load	**VT; surface profilometry; cross-section metallography to reveal subsurface void**	Extend burn-off stage; reduce heating rate through 300–500°C; verify lubricant content and blend uniformity
Dimensional Variation	Batch-to-batch changes in powder characteristics (particle size, apparent density) shift green density and sintering shrinkage (1–3%) — final part dimensions change between batches	Parts outside dimensional tolerance — not detectable by visual inspection; found only by measurement	All PM parts; most critical on precision bores, mating faces, and any feature with tight tolerance	**CMM or gauge measurement; 100% dimensional inspection is standard for critical PM components**	Tighten powder characterization specification; coining or sizing after sintering to correct critical dimensions

SLIDE 12 *PM Defects — Recognition and Detection*

NOTES

Manufacturing Process Selection — Unit 4 Summary

5-Step Decision Framework

1 Alloy compatibility
Does the alloy constrain the process? Die casting cannot process steel; PM cannot easily process reactive alloys without vacuum sintering; investment casting handles almost any alloy.

2 Part geometry
Very large parts eliminate die casting and PM; complex internal geometry favours casting with cores; parts with tight bores and flat surfaces require machining.

3 Tolerance & surface finish
Sand casting for loose tolerances; grinding for very tight (Ra 0.025–0.8 μm); PM for moderate; machining bridges the gap between casting and grinding.

4 Production volume
High volume favours die casting, PM, and cold forming with high tooling investment; low volume favours sand casting and machining from bar stock.

5 Cost target
PM and die casting amortise high tooling cost over many parts; sand casting and machining have low tooling cost but higher per-part cost.

Unit 4 Process Comparison

Casting
Alloy: excellent flexibility. Geometry: complex external and internal (cores). Volume: sand = low; die = high. Finish: rough to moderate. Key defect: porosity.

Forming (Forging & Rolling)
Alloy: wrought metals only. Geometry: limited — no internal features. Volume: medium to high. Finish: moderate. Key advantage: superior grain flow and fatigue life.

Machining
Alloy: almost any. Geometry: any external and internal features. Volume: any. Finish: finest (Ra 0.025–0.8 μm). Limitation: material waste 50–90%; cost is high per part.

Powder Metallurgy
Alloy: limited (no reactive without vacuum). Geometry: moderate — vertical ejection only. Volume: medium to high. Key niche: WC-Co tools, self-lubricating bearings, near-net-shape.

SLIDE 13 *Manufacturing Process Selection — Unit 4 Summary*

NOTES

Core Question Answers

What is powder metallurgy and what advantages does it offer over casting and machining?

PM starts with metal powder pressed into a die at 150–700 MPa then sintered at 70–90% of the melting point — solid-state diffusion bonds particles without melting. Advantages: near-net-shape with minimal material waste; enables materials impossible by other routes (WC-Co cutting tools, self-lubricating porous bearings, controlled-porosity filters); eliminates many secondary operations. Limitation: 85–95% theoretical density; economical only at moderate to high volumes due to tooling cost.

How are metal powders produced, compacted, and sintered?

Production: water atomisation (irregular, iron/steel), gas atomisation (spherical, stainless/superalloys), reduction (sponge iron), electrolytic/carbonyl (high purity). Compaction: blended with lubricant, die-filled, pressed from both ends at 150–700 MPa — green compact has 5–15 MPa strength. Sintering: burn-off lubricant at 300–500°C, neck formation at 70–90% Tm (~1100–1150°C for iron), densification with controlled atmosphere (H_2, N_2, or vacuum) and predictable 1–3% shrinkage.

What are the limitations of PM and what defects should a technician look for?

Limitations: density 85–95% theoretical; geometry limited to vertical die ejection; tooling cost high; part size limited by press capacity. Key defects: density gradients (uneven compaction — detect by hardness traverse or UT); lamination cracks (spring-back on ejection — planar crack perpendicular to pressing axis, detect by UT); blistering (rushed burn-off stage — VT); dimensional variation (powder batch changes — 100% measurement inspection).

How do you select the right manufacturing process for a given application?

Work through five factors in sequence: (1) alloy compatibility — does it constrain the process? (2) geometry — does part complexity eliminate options? (3) tolerance and surface finish required; (4) production volume — high volume justifies high tooling investment; (5) cost target. Most finished parts involve multiple processes in sequence — a casting that is then machined, or a forging that is then ground. Understanding the full manufacturing chain, not just the individual operation, is the mark of a quality technician.

SLIDE 14 *Core Question Answers*

NOTES

Key Takeaways — Week 13: Powder Metallurgy

1 Powder metallurgy occupies a unique manufacturing niche — it is the only practical route to certain materials (WC-Co cutting tools, self-lubricating bearings, controlled-porosity filters) and offers exceptional material efficiency; its limitations in density and geometry constrain where it can be used but do not diminish its importance in the manufacturing landscape.

2 Sintering is a thermally activated diffusion process, not melting — controlling atmosphere, temperature, and time determines the final density, microstructure, and mechanical properties of the PM part; deviations from the qualified sintering cycle are a quality concern just as critical as incorrect heat treatment in steel.

3 Process selection across all Unit 4 processes requires simultaneously satisfying alloy, geometry, tolerance, volume, and cost requirements — no single process is universally best, and most finished components involve multiple processes in sequence; a technician who understands the capabilities and limitations of each process can reason about the entire manufacturing chain, not just the individual step they are inspecting.

SLIDE 15 *Key Takeaways — Week 13: Powder Metallurgy*

NOTES

Resources & What's Next

Week 13 Resources

Kalpakjian Ch. 17
Powder Metallurgy — production, compaction, sintering, and applications — primary reading

MPIF Process Overview
Metal Powder Industries Federation interactive PM process visualizer

Sintering — Neck Formation
Animated sintering stages: burn-off, neck formation, and densification

Atomisation & Powder Production
Water vs. gas vs. reduction methods — particle shape and characteristics

Manufacturing Process Selection
Unit 4 summary tool — casting vs. forming vs. machining vs. PM

Week 14 — Welding Processes

November 23, 2026 · Unit 5: Joining & Failure begins

- Fusion welding processes — SMAW, MIG, TIG, and their applications
- Heat input and the weld thermal cycle — heat-affected zone formation
- Weld joint design — groove, fillet, plug, and slot welds
- Solidification of the weld pool — columnar grain growth and cracking
- Weld inspection and qualification — VT, RT, UT, MT, and PT

SLIDE 16 *Resources & What's Next*

NOTES